Practical 1
Title - Performing the analysis using excel

Theory –
Data Warehouse
A data warehouse is the foremost repository for the data available for developing business intelligence architectures and decision support systems. The term *data warehousing* indicates the whole set of interrelated activities involved in designing, implementing and using a data warehouse. It is possible to identify three main categories of data feeding into a data warehouse: *internal data*, *external data* and *personal data*.
Data marts
Data marts are systems that gather all the data required by a specific company department, such as marketing or logistics, for the purpose of performing business intelligence analyses and executing decision support applications specific to the function itself. Therefore, a data mart can be considered as a functional or departmental data warehouse of a smaller size and a more specific type than the overall company data warehouse. A data mart therefore contains a subset of the data stored in the company data warehouse, which are usually integrated with other data that the company department responsible for the data mart owns and deems of interest. For example, a marketing data mart will contain data extracted from the central data warehouse, such as information on customers and sales transactions, but also additional data pertaining to the marketing function, such as the results of marketing campaigns run in the past.

a. **Import the data warehouse data in Microsoft Excel and create the Pivot table and Pivot Chart.**
Procedure –

Creating Data Warehouse
Step 1: Download T-SQL script attached with this article for creation of Sales Data Warehouse or download from this article "Create First Data Warehouse" and run it in your SQL Server.
Downloading "Data_WareHouse__SQLScript.zip" from the article
https://www.codeproject.com/Articles/652108/Create-First-Data-WareHouse
Follow the given steps to run the query in SSMS (SQL Server Management Studio).
1. Open SQL Server Management Studio 2012
2. Connect Database Engine
3. Open New Query editor
4. Copy paste Scripts given below in various steps in new query editor window one by one
5. To run the given SQL Script, press F5
6. It will create and populate "Sales_DW" database on your SQL Server.
7. After completing execution save and close SQL Server Management studio & Reopen to see Sales_DW in Databases Tab.

Creating Pivot Table and Creating Pivot Chart
Step 1: Open Excel
Go to Data tab → Get External Data → From Other Sources → From Data Connection Wizard
Step 2: In Data Connection Wizard → Select Microsoft SQL Server → Click on Next
Step 3: In connect to Database Server provide Server name(Microsoft SQL Server Name)
Step 4: In Select Database and Table→ Select Sales_DW (already created in SQL) → check all dimensions and import relationships between selected tables
Step 5: In save data connection files browse path and click on Finish
Step 6: In import data select Pivot Chart and click on OK

Step 7: In fields put SalesDateKey in filters, Product Name in axis and Sum of ProductActualCost in values
Step 8: In Insert Tab → go to Pivot Table
Step 9: Click on Choose Connection to select existing connection with Sales_DW and click on open Pivot table and Pivot chart is created